

Cerium

58
Ce

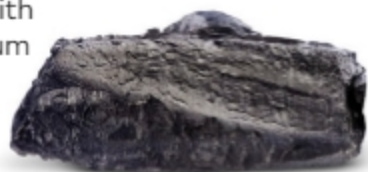
In 1803, cerium was discovered and named after the dwarf planet Ceres (which itself was named after the Roman goddess of agriculture) by one of its discoverers, Swedish chemist Jöns Jakob Berzelius. An important use of cerium is in white LEDs, and it's also used to make some fade-resistant pigments.

Laboratory sample of pure cerium

Praseodymium

59
Pr

A part of praseodymium's name comes from the Greek word *prasios*, meaning "green," because it turns green when it tarnishes in air and forms other bright green compounds. Combined with neodymium, praseodymium is used to create high-power magnets.



Laboratory sample of pure praseodymium

These glasses from the 1920s contain praseodymium that makes them green.

Filter for safety glasses

Praseodymium is used to give a colored tint to the safety glasses used by welders and glassblowers. The tinted goggles filter out the potentially damaging ultraviolet radiation, protecting the eyes of the wearer.



Samarium

62
Sm

This element was discovered by French chemist Paul-Émile Lecoq de Boisbaudran and named after the mineral samarskite from which it was first extracted. Samarium is added to cobalt to make powerful magnets that are useful because they continue to work at high temperatures.



Laboratory samples of pure samarium

Europium

63
Eu

Paul-Émile Lecoq de Boisbaudran found europium first in 1892, but its discovery is usually credited to the French chemist Eugène-Anatole Demarçay, who isolated it in 1901. It's used in the inks used to print euro banknotes—under ultraviolet (UV) light, the notes glow red, proving they are genuine.

Laboratory sample of pure europium

